Chapter 5
The internet and its uses

IN THIS CHAPTER YOU WILL:

- learn what the difference is between the internet and the world wide web
- understand what is meant by a URL and study the purpose of each of its component parts
- discover the purpose and operation of the hypertext transfer protocol (HTTP) and the hypertext transfer protocol secure (HTTPS)
- learn about the different functions that a web browser performs, including the use of cookies
- discover how web pages are requested, retrieved and displayed on your computer or device
- understand what is meant by a digital currency and how blockchain is used to track digital currency transactions
- learn about a range of cyber security threats and how a range of different solutions can be used to keep data safe from these threats.

GETTING STARTED

Think about what you already know about keeping yourself safe on the internet (e-safety). Create a poster for a younger audience to tell them how to use the internet safely and what to do if they find something that upsets them.

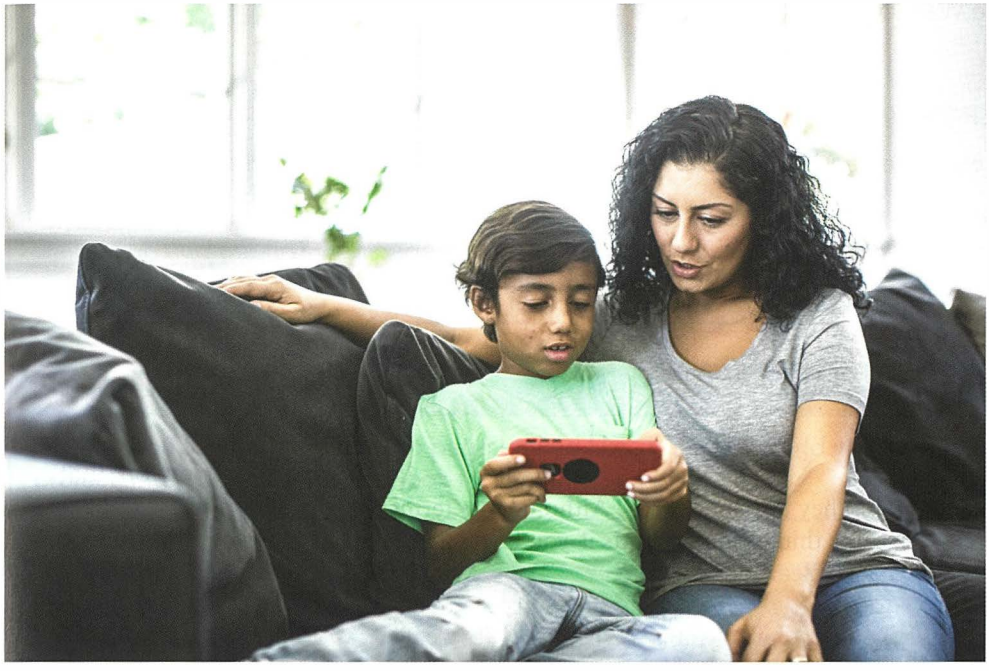


Figure 5.1: A parent helping a child to safely access the internet

THE EFFECTS OF MALWARE

On average, there are approximately 2500 cyber security attacks that happen each day. Malware is used in many of these attacks and over 90% of this malware is delivered using email.

WannaCry is the name of a large scale cyber security attack that occurred in 2017. Malware was used to take over computer servers in systems across Britain, Spain, Russia, Ukraine and Taiwan. The type of malware used in the WannaCry attack is called ransomware. The malware blocked users and organisations from accessing files and accounts. It did this by encrypting the files. It wanted the organisations to pay money to the creators of the malware in order to be able to gain access to their files and accounts again. The payment demanded was in Bitcoin.

As organisations were not able to gain access to their files and accounts until the ransom was paid, mass disruption was often caused to the

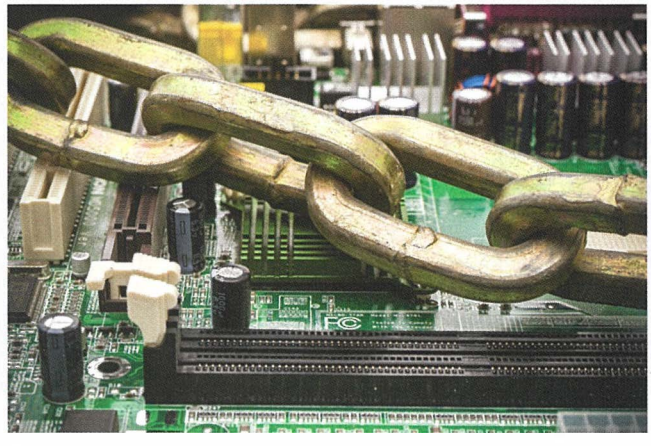


Figure 5.2: Malware prevents users from accessing their files and data

CONTINUED

daily processes in these organisations. One example of this was the National Health Service (NHS) in Britain. British hospitals were forced to turn away some patients who could not be treated until their personal information could be accessed again.

WannaCry malware was based on a set of malware code called EternalBlue. There have been several malware attacks since WannaCry based on this same EternalBlue code.

Discussion questions

1 Do you think it is the organisation's fault that the malware infected their systems and caused issues?

2 Do you think it is completely the hacker's fault for being able to put the malware on the systems?

3 How would you feel if your files were encrypted and you couldn't access them? Would you pay the ransom, or would you take a different action?

5.1 The internet and the world wide web

The internet and the world wide web: aren't they the same thing? Although the terms are used interchangeably, they aren't the same thing. Bonus points if you knew that already! The **internet** is an **infrastructure**. This means it is all the components and cables that are used to connect one device to another. Another way that the internet can be described is that it is one huge **network**. More specifically, it is a type of network called a wide area network (WAN). This is a type of network that spans a large geographic area, in this case the whole world! The internet is an incredible structure if you think that homes and businesses all over the world are able to use it to connect to each other. Millions and millions of people use this huge infrastructure.



Figure 5.3: The internet connects millions of people

The **world wide web** is the collective term for all the **websites** and **web pages** that are available. The world wide web is accessed by using the internet.

KEY WORDS

internet: a very large global network that allows users to access the world wide web.

infrastructure: the physical structure, such as all the components and cables, that are used to connect devices.

network: computers and devices that are joined together using cables or wireless technology.

world wide web: the collection of all the web pages and websites that have been made available.

website: a collection of web pages, normally about a related topic or purpose.

web page: a single page within a website.

COMPUTER SCIENCE IN CONTEXT

The world wide web was invented by Tim Berners-Lee in 1989. He was motivated to do so by a problem that he had. He is a Computer Scientist and found that at the time he had several computers that all had different information stored on them. He found that he would have to keep logging onto the different computers to obtain the particular information that he needed. He saw a way to solve this issue that he realised could be used universally. Lots of computers were already being connected together into networks and he realised that he could share information using an emerging technology at the time called Hypertext. By October of 1990 he had developed the three fundamental parts that create the world wide web: HyperText Markup language (HTML), uniform resource locator (URL) and the Hypertext Transfer Protocol (HTTP). By the end of 1990, the first web server was set up and the first web page displaying information about the world wide web project was created.

ACTIVITY 5.1

Have a look at what the first web page looked like. Type 'the world wide web project' into your browser and look for the 'info.cern' link.



Figure 5.4: HTML

5.2 Requesting and retrieving web pages

A **URL** is a fundamental component of the world wide web. Each website and web page on the world wide web has a text based address. This address is called a URL. To access the web page, a user types the URL into the address bar of a **web browser**.

A URL has three main components: the **protocol**, the **domain name** and the web page or file name.

KEY WORDS

uniform resource locater (URL): another name for the text-based address for a website.

web browser: a piece of software that retrieves and displays web pages.

protocol: a set of rules for transmitting data.

domain name: a specific part of a URL that contains the unique name for the web server.

Figure 5.5 shows an example of a URL:

https://www.cambridge.org/education

protocol | domain name | web page name

Figure 5.5: The components of a URL

There is a whole process involved in requesting and retrieving web pages that you need to understand, the URL and its three components are fundamental to this process. So how is a web page requested and retrieved by our computers?

To start the process, the user opens their web browser and types the URL into the address bar. The web browser then sends the URL for the website to the **domain name server (DNS)**. The DNS is a special kind of server that stores all the different domain names (part of the URL shown in Figure 5.5) along with their equivalent IP address. The DNS looks through its database of domain names to find the equivalent IP address. As you can imagine, there are billions of websites, so these are not all stored on a single DNS. Therefore, the first DNS will search its database and if it doesn't find the domain name, it sends it to the next DNS. This will happen until the domain name is found, or if it is not found, a message is sent back to the web browser to say that the website is not found. When the DNS find the domain name, it sends the equivalent IP address back to the web browser – the IP address of the **web server** that stores the website. This process is summarised in Figure 5.6.

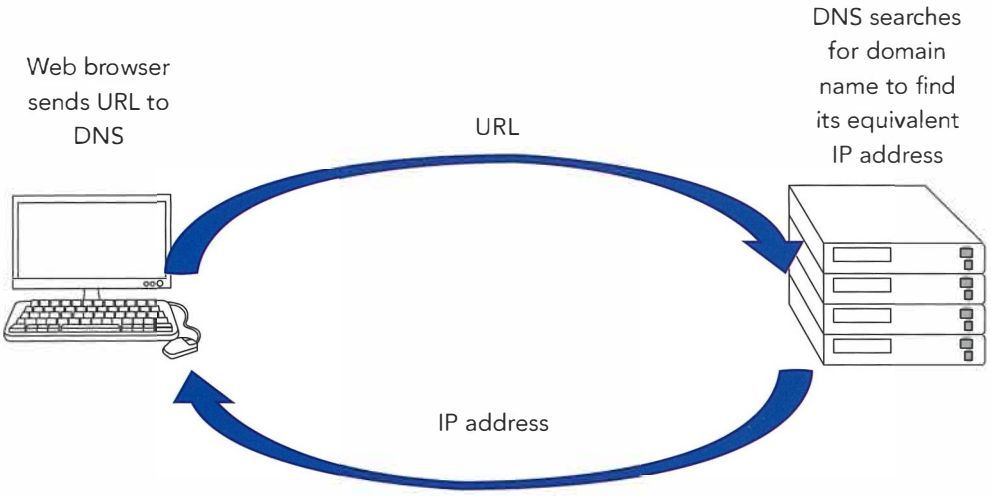


Figure 5.6: A web browser requests an IP address from a DNS

The web browser receives the IP address for the website from the DNS. It now knows where to find the website. The web browser sends a request to the web server to ask for the web page from the website. It uses a protocol called the **hypertext transfer protocol (HTTP)** to send the request to the web server. Each web page is created using **hypertext markup language (HTML)**, **cascading style sheets (CSS)** and may include **active script** such as JavaScript. When the web server receives a request for a web page, it sends the data for the website, including the HTML, CSS and any active script, back to the web browser using the HTTP. The web browser then renders the HTML and CSS and runs any active script to display the web page. This process is summarised in Figure 5.7.

KEY WORDS

domain name server (DNS): a special server that contains a database of domain names and their corresponding IP address.

web server: a network component where the web pages for a website are stored.

hypertext transfer protocol (HTTP): a protocol that is used for the transmission of web pages and related data across the internet.

hypertext markup language (HTML): a scripting (web authoring) language that is used to create web pages.

cascading style sheet (CSS): a scripting language that is used to create a presentation template for a web page. It includes what kind of font and colour text will appear on the webpage.

active script: this is the script that is used to produce any of the interactive elements of a web page.

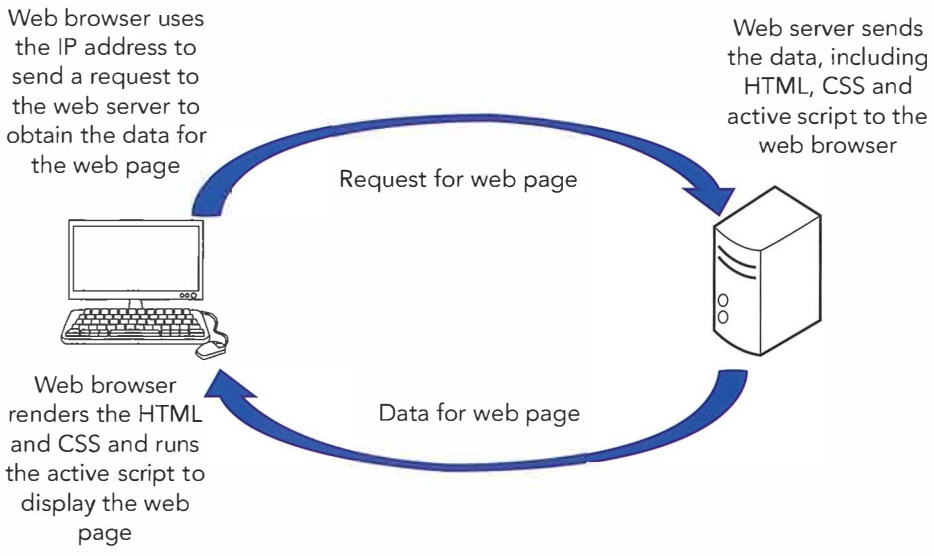


Figure 5.7: The web browser requests a web page and renders the HTML and CSS

Questions

1 What is a URL?

2 What is an IP address?

3 What are the **three** main parts of a URL?

Another protocol can be used to request the data for the web pages that will require an additional process to take place. This protocol is the **hypertext transfer protocol secure (HTTPS)**. This protocol has an additional layer of security that **encrypts** any data that is transferred. It involves the use of **digital certificates**. This is a certificate that the owners of the website need to apply for from a **certificate authority**. A certificate authority is an organisation that checks whether a website is authentic and trustworthy. If the certificate authority believe the website is authentic, they will issue them with a digital certificate. This is how the additional layer of security is created.

- Before the web browser sends a request to the web server to ask for the web pages, it sends a request to the web server asking it to provide a digital certificate.
- The web server sends a copy of the digital certificate to the web browser.
- The web browser checks if the digital certificate is authentic.
- If the certificate is authentic, the web browser will allow communication to continue with the web server and any data that is transmitted between the two is encrypted.
- If the certificate is not authentic, the web browser will report that the website is not secure.

KEY WORDS

hypertext transfer protocol secure (HTTPS): a secure version of the HTTP that encrypts data for transmission.

encryption: a method of securing data for storage or transmission that scrambles it and makes it meaningless.

digital certificate: a certificate that is awarded to a website if they can prove that they are a real organisation and take measures to securely transmit their data.

certificate authority: the awarding organisation that checks if another organisation is real and secure before awarding them a digital certificate.

This process is summarised in Figure 5.8.

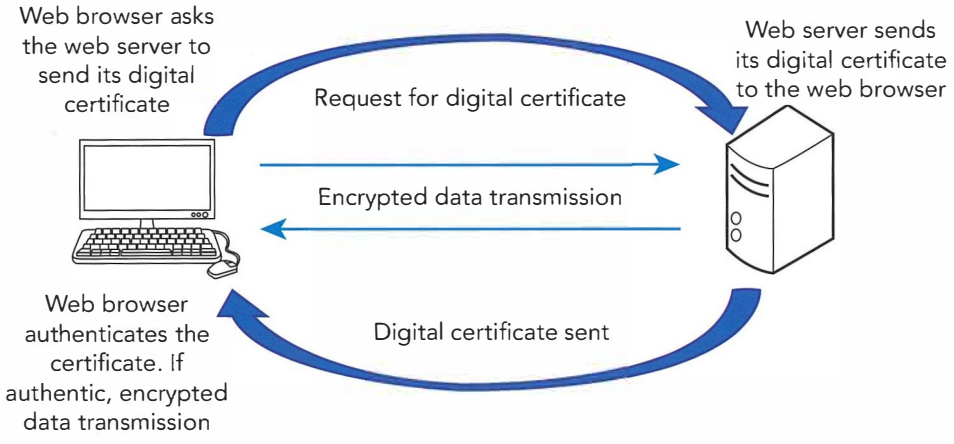


Figure 5.8: A web browser requests a digital certificate to allow encrypted data transmission

The HTTPS protocol is the standard HTTP protocol with a layer of security combined, that is either the **secure sockets layer (SSL)** or **transport layer security (TLS) protocol**. TLS is a newer version of SSL. This is the protocol that is used to create the encryption. You can check visually whether a website is secure by seeing if it uses the HTTPS protocol. If it does, you will see this at the start of the URL in the address bar in the web browser. You will also see a small padlock before the start of the URL. Most web browsers also have an option for you to view the digital certificate for the website. This is often found by right clicking your mouse on the padlock.

ACTIVITY 5.2

Use the internet to research why the TLS protocol was developed to replace the SSL protocol.

Questions

4 What is a digital certificate and how is it used in the HTTPS protocol?

5 Does the web browser or the web server authenticate the digital certificate?

KEY WORDS

secure sockets layer (SSL) protocol: a type of protocol that encrypts data for transmission. It is the protocol, that is combined with the HTTP to create HTTPS.

transport layer security (TLS) protocol: an updated version of the SSL protocol.

render: the method of processing all the web page data, such as HTML, to display the web page.

5.3 The purpose of a web browser

You may have noticed that the web browser has a key role in the requesting and retrieving of web pages. The main purpose of a web browser is that it is a software application that allows you to access information that is available on the world wide web. As you learnt previously, it does this by requesting the HTML files and other data required to create the web page, then **renders** this data so that you can view the web page.

One thing to note about a web browser is that some people confuse it with a search engine. This is because some web browser software development companies also

produce search engines too. One example of this is Google. Google produce a search engine called google, that you can use to find information on the world wide web, however this is not a browser. The browser that Google produce is Google Chrome. This is the software that allows you to open web pages, such as the google search engine. Why don't you test your friends and see if they know the difference?



Figure 5.9: Example of a search engine

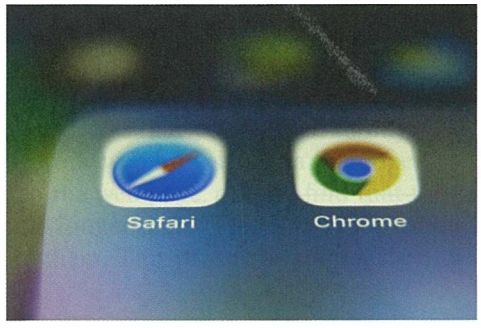


Figure 5.10: Examples of web browsers

ACTIVITY 5.3

Can you think of two more examples of a web browser? Use the internet to check if you are correct.

Did you know that there are lots of other things that a web browser can do? This is a list of some of the main functions of a web browser, how many have you used before?

- It provides an **address bar** that you can use to type in a URL.
- It keeps a record of all the web pages that you have visited. This is called **user history**.
- It allows you to **bookmark** web pages and set web pages as your favourites. These can then be displayed on the toolbar of your web browser to allow you to access them much more quickly. It means you can just click the bookmark or favourite icon and it will take you straight to the web page.
- It allows you to open multiple **tabs** at a time. The browser itself opens into a **window** on your computer. Without tabs, you would need to open multiple windows if you wanted to open more than one web page at a time. Tabs allow you to open multiple web pages in the same browser window which makes it easier to navigate between the different web pages that you have open.
- It allows you to **navigate** between web pages. The buttons that are most commonly available on a web browser are forward, back and reload the page.
- It stores **cookies** and manages the process of transmitting them. You will learn more about cookies next!

KEY WORDS

address bar: the section of a web browser where you type the web page address.

user history: a record made on your web browser of all the web pages that you have visited.

bookmark: a link that can be created to allow you to quickly find a web page again.

tab: a website function that allows you to open multiple web pages in the same window.

window: a viewing area that is created in which a software application or file can be opened.

navigate: move around web pages, or move back or forward to previous web pages.

cookie: small text file that is used to store personal data, by a web browser.

The main features are shown in Figure 5.11.

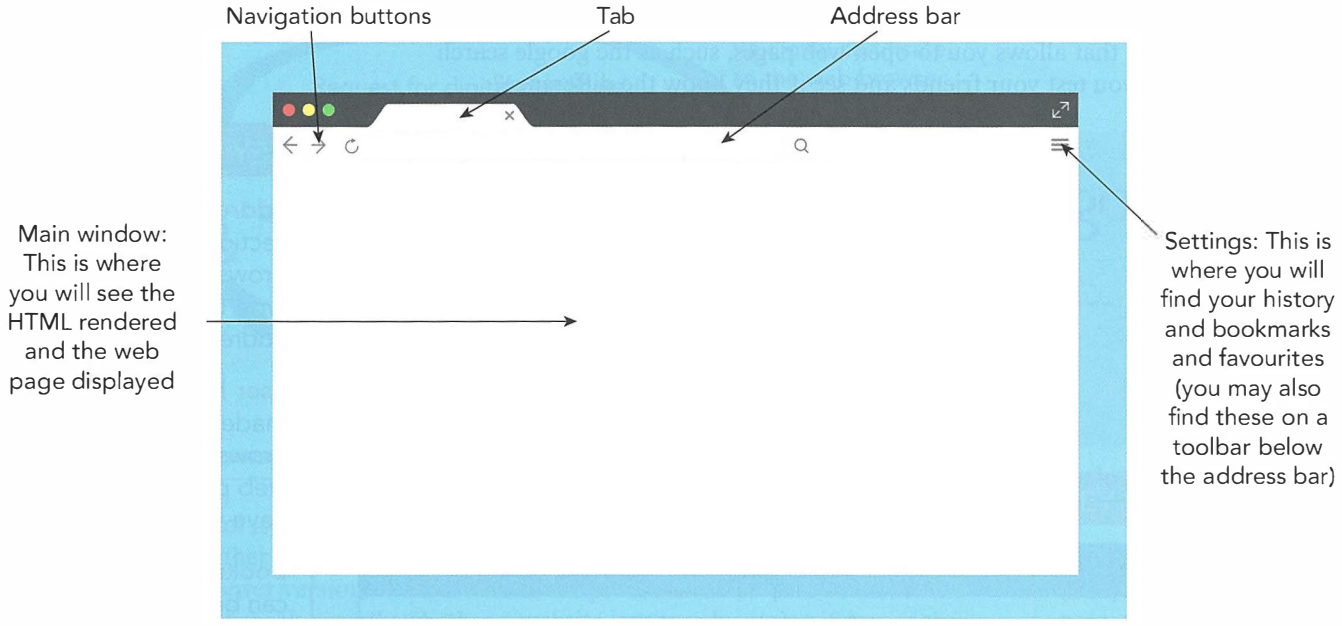


Figure 5.11: The main features of a web browser

Cookies! Sadly, not the tasty kind, but a useful function of a web browser. So, what exactly is a cookie? A very simple explanation is that it is a small text file that is used to store data, normally about you.

They are very useful as they can store data that you may regularly use on the internet. This means that you do not need to enter this data every time. Examples of this are:

- Storing your username and password for accounts such as your social media.
- Storing your bank card details for buying products and services online.
- Storing products in a virtual shopping cart when you are buying multiple products from an online company. It can also save the items in your cart so that you can buy them at a later date if you don't want to buy them now.

Cookies can be used to tailor your online experience with websites. Some websites allow you to format the website in the way that you want it to look, for example, changing the colour of text and boxes that appear on the web page. Cookies can also be used to track the different content that you look at on the world wide web. For example, if you look at web pages about your favourite hobby or interest, you will find that advertisements appear on web pages that are about your favourite hobby too. This is called targeted advertising.

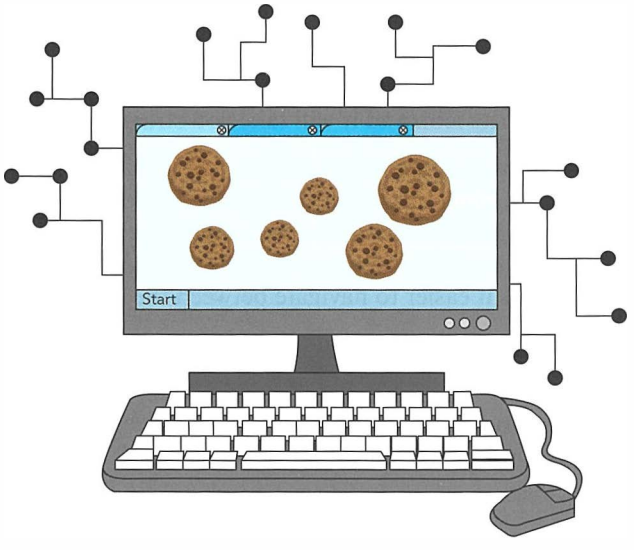


Figure 5.12: Many websites use cookies

Cookies are created when you visit a web page and then they are stored by your web browser. There are two main types of cookie, **session cookies** and **persistent cookies**.

Session cookies are temporary files that are created when you visit a web page. As soon as you close your web browser, these cookies are deleted. If you open your web browser again and go back to the same web page, the web page will not recognise you. That means that it will not be sent any cookies by your web browser, about your personal details or preferences, as this type of cookie are not saved by the browser once it is closed. You might have experienced this type of cookie when you are online shopping. You may have added several items that you wanted to buy to a shopping cart on a website, you then close your web browser. You then open your web browser at a later date and go back to the web page, only to find that the items you put in the online shopping cart are no longer there. This is because it is a session cookie that is used to temporarily store them. This process is summarised in Figure 5.13.

KEY WORDS

session cookie: a type of cookie that is deleted when the web browser is closed.

persistent cookie: a type of cookie that is stored by the web browser until it is deleted by the user or because it has expired.

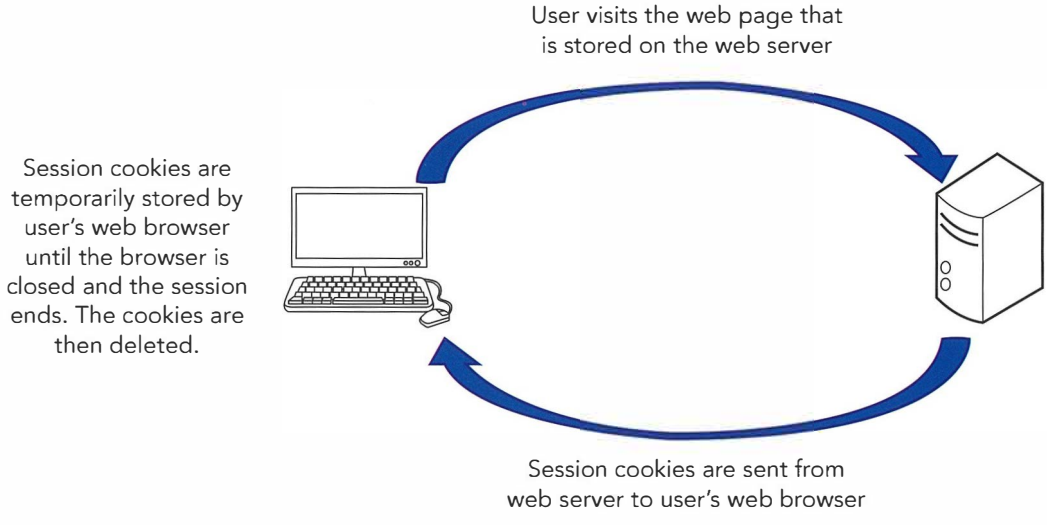


Figure 5.13: Session cookies

Persistent cookies are permanent files that are created when you visit a web page. These are stored by your web browser onto the hard drive of your computer. When you visit the same web page again, the cookie file is sent back to the web server that stores the web pages to provide the data about your personal details or preferences. This will allow your login details to be automatically entered, for example. These cookies will remain stored on your hard drive until you manually delete them, or your browser deletes them because they have expired. Each persistent cookie file is sent with an expiration date. When this expiration date is reached, the web browser will delete this cookie from your hard drive. This process is summarised in Figure 5.14.

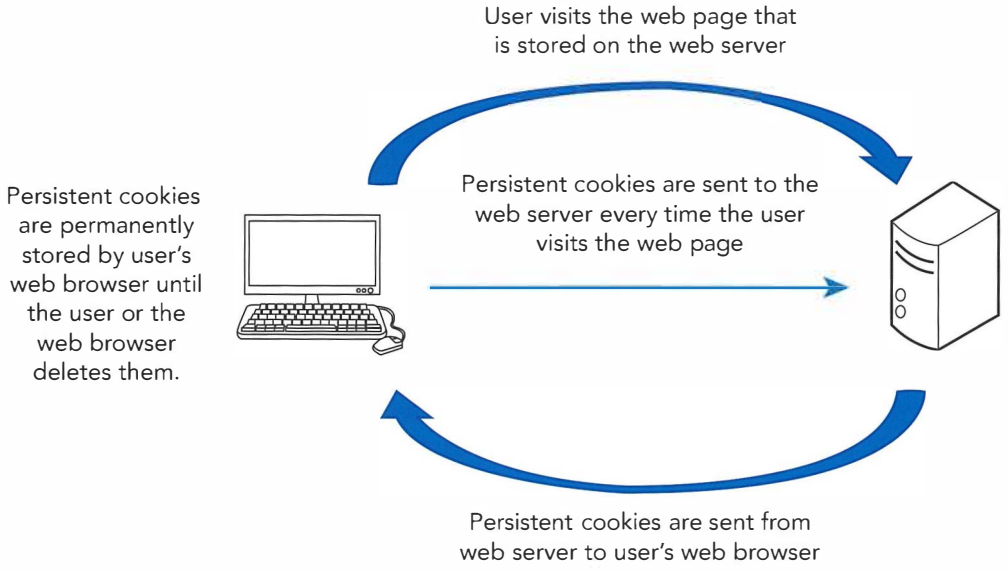


Figure 5.14: Persistent cookies

Internet users have mixed feelings about cookies. This is because they can have a very useful role but can also be used for improper activities. Cookies do build up an online profile about your details and preferences. Some people feel this is an invasion of their privacy. Other people also believe that this data can be gathered to build a profile about you and add to the risk of your identity being stolen. If you are worried about this being a possibility, you can limit the use of cookies in your online activity by changing the settings for your web browser.



Figure 5.15: Cookies can be a privacy concern

Questions

6 What is a cookie used for?

7 What is the difference between a session cookie and a persistent cookie?

8 Why do some people have an issue with cookies being used?

COMPUTER SCIENCE IN CONTEXT

Each company that creates a web browser will often have a slightly different aim as its focus. Some web browser development companies focus on the security of the transmission of data, whereas others focus on how quickly the web pages are rendered and displayed.

5.4 Digital currency

When you pay for products and services using the internet, you will normally use a **digital currency**. A digital currency is one that exists electronically (see Figure 5.16). It is a method of payment that is similar to paying with coins and bank notes. However, rather than exchanging physical bank notes and coins to make the payment, the currency is exchanged digitally using computers. There are several different forms of digital currency, the most popular are payments using credit cards, mobile phones and smart watches. The most common of these is using credit cards when making payments using the internet.

KEY WORD

digital currency: a currency that exists electronically rather than physically.

When the data about the payment with the digital currency is sent from one computer to another, it is encrypted to keep it secure. The method of encryption used is what you learnt about previously, in Section 5.2, using the HTTPS protocol. The payment details for the use of a digital currency can be stored in a persistent cookie. That means that each time you visit the website to buy a product, your payment details will be automatically entered so you don't need to type them in each time.



Figure 5.16: Digital currency exists electronically and can be sent between computers

One type of digital currency that you may have heard of is Bitcoin. This type of digital currency is more specifically known as a **cryptocurrency**. The banks that we use to manage our money on a daily basis are called centralised systems. This means that there is an authority at the centre managing the process in which they are used. In this case, the bank is the authority. A cryptocurrency is a decentralised system, this means that there is no central authority, like a bank, managing the process of payments. Payments are also encrypted using this type of currency. As there is no central authority managing the system of payments with this type of digital currency, a different system called **blockchain** is used to keep track of the payments.



Figure 5.17: A cryptocurrency is a digital currency that is managed by a decentralised system

KEY WORDS

cryptocurrency: a type of digital currency that uses encryption procedures.

blockchain: a method that is used to track all the transactions made with a cryptocurrency.

digital ledger: a database that is a list of all the transactions recorded by the use of blockchain.

In its most basic form, blockchain is a list of all the records made with the digital currency. This is called a **digital ledger** (Figure 5.18). Each time a payment is made with the digital currency, a record is added to the ledger that includes a digital signature with the time and date that the payment is made. In blockchaining, once these records are added to the digital ledger, they cannot be changed. Data that is stored in a digital

ledger is also encrypted. This prevents anyone being able to change the data. However, anyone who wants to view the data in the digital ledger is able to do so.

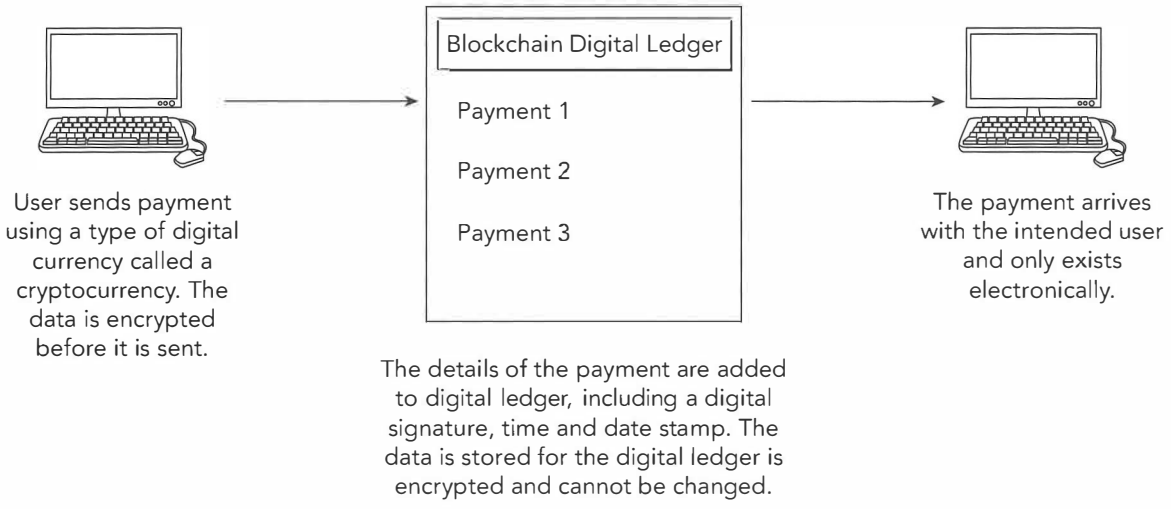


Figure 5.18: A blockchain keeps a digital ledger of transactions

COMPUTER SCIENCE IN CONTEXT

There are as many as 2000 different digital currencies that exist. Many of these are different types of cryptocurrency. How many do you think you could name?

5.5 Cyber security

Whilst the internet and the world wide web can be a very fun and interesting place, it does also have some issues and threats that you definitely need to be aware of. You need to know how these cyber threats are created, what the motivation is behind those who use them, and what kind of security solution you can put in place to try and prevent them from happening. One thing to note is that each security solution may not be a complete prevention, sometimes it is better to have several solutions in place. Also, **perpetrators** are always looking to develop ways to beat security solutions, so no solution will be 100% effective in preventing an attack. However, if you don't have any security solutions in place, you are putting your personal data at a far greater risk of these cyber threats. So, what are the cyber threats that you need to know?

KEY WORDS

perpetrator: a person that carries out an illegal or immoral action.

brute-force attack: a type of cyber threat that involves repeatedly inputting a password until the correct one is found, to try and break into an account or device.

Brute-force attack

You may have heard someone being described as a brute person. This normally means people think they are strong and forceful. This is exactly what a **brute-force attack** is on your computer, a strong and forceful attack. Imagine that someone is banging on a locked door, pushing and shoving it in every way that they can, until the door gives way and opens. In a brute-force attack, a perpetrator tries to break into your computer or account. They do this by repeatedly entering different password combinations until they manage to enter the correct one. They may try to do this manually, by typing in each password combination themselves. They could also do this by developing a

software program that will automatically cycle through many different combinations until the correct one is found.

The perpetrator usually does this to gain access to your computer or account in the hope that they can either steal some personal data that they can use for criminal activity, or they may try to use your account to purchase items online as though it was their own account.

There are several cyber security solutions that you can put in place to help prevent a brute-force attack on your computer or account. The first is to make sure that you have a strong password to protect your data. A strong password is created by mixing different uppercase and lowercase letters, numbers and symbols. Ideally mixing these in a way that does not form any kind of word or phrase. A long string of a mixture of all these characters is the best protection that you can give your data. This will make it much more difficult for any perpetrator or software program to guess your password. It can be easy to find out some simple information about you, for example, that you have a rabbit called Mr Snuffles. Therefore, if your password is mrsnuffles01, this is not going to be difficult for a perpetrator to guess. Make life a bit more difficult for them, your data is precious! An example of a strong password is:

1gH6dk@dnlwm!dosTsjlpa

Many web browsers now have a function that will suggest a strong password for you, mixing a combination of characters like this. They will also save the password for you, so you don't need to keep remembering the string of characters.

Another security solution that can be used to help prevent a brute-force attack is limiting the number of times a password can be entered before the account is locked. You may have had this issue before. You type in what you think you had as a password for the account, but it is rejected, so you try another and that gets rejected, then you try a third one and that is rejected, and you now get a message saying you are locked out of your account. How frustrating! However, be very thankful for this security solution, however annoying it can be. It is likely that for a perpetrator to brute-force their way into your account, it will take more than three guesses at your password. Therefore, if the number of attempts is limited before the account is locked, this can prevent many guesses being made to try and find out the password.

A third security solution that can be used is a **biometric password**. A **biometric device** is one that needs your biological data to be input. An example is a fingerprint scanner, which requires your fingerprint to be input. Our biological data is unique to us, it is highly unlikely that you will find another person that has exactly the same fingerprint as you. Therefore, if you secure your computer or account with a biological password, such as your fingerprint, no amount of guessing your password would allow a perpetrator to brute-force their way into your account. They would need to somehow obtain your fingerprint, which is a much more difficult task.

A fourth security solution that can be used is **two-step verification** (sometimes known as two-factor authentication). This is a method that puts two tasks in place in order to enter an account. The first task is entering the password, but then a second step is added that will require an additional input task to enter the account. The second piece of data is often a code or word that is sent to you that you will need to enter. This is normally sent to either your mobile phone number or your email address. This means that a perpetrator will not only need to guess the password for your account, but that they will also need to have access to your mobile phone or your email, this is an added difficulty. Some banks also have a small card reading device that provide you with the data for the second step. You put your bank card into the reader, enters a PIN code

KEY WORDS

biometric password: a type of password that uses a person's biological data, such as their fingerprint.

biometric device: a device that allows a user to record and input a biometric password.

two-step verification: a process that involves inputting two different kinds of data to enter an account or device.

or password and this generates the code for the second step for you to input. It is advisable that you look at the privacy and security settings for your online accounts, especially ones like your email and social media. Many companies have an option for you to turn on the feature of two-step verification which helps keep your personal data more secure.

In summary:

What happens?	Multiple guesses are made at your password to try and break into your computer or account.
What is the aim?	To steal your precious personal data or to use your account to buy products online.
What security solutions can be used?	A strong password, a limited number of attempts at entering a password, a biometric password and two-step verification.

ACTIVITY 5.4

Think about one of your passwords that you use to access an account online. How strong do you think that password is? Do you think that you could improve that password?

Create a three-slide presentation that can be given to your peers to tell them how to create a strong password and what the importance is of doing so.

Peer Assessment

Take a look at your partner's presentation. Do you think the information is presented clearly? Have they made it clear how to create a strong password? Do you think they have missed any information about doing this? Do you think they have made it clear how important it is to set a strong password? Have they included information about what might happen if a person does not set a strong password?

Tell your partner your answer to these questions when you have looked at their presentation.

Data interception

Billions of pieces of data are sent across the large network that is the internet on a daily basis. A lot of this data is people sending their personal data to do things like buy products and services, and log into their online accounts. Perpetrators would very much like to get hold of their information so that they can use it in criminal activity, such as identity theft and fraud. A way that they will try and do this is by intercepting **data packets** as they are transmitted across the network from one device to another. This is often done by using a piece of software called a **packet sniffer**, this is normally installed onto a piece of hardware that the packets will pass through, for example, a router. This software is used to examine packets of data that are sent across a network to see if they contain data that looks useful. If they do, the software will report the useful data packets it has found back to the perpetrator.

KEY WORDS

data packet: a unit of data that is used to transmit data across a network.

packet sniffer: a piece of software that is used to examine the contents in a packet of data.

The main security solution that you can use to keep data secure that is sent over a network is encryption. Data is normally encrypted using the SSL protocol that you learnt about earlier in Section 5.2. This security solution will not stop the data packets from being intercepted, but it will mean that if the data packets are intercepted, the data contained in the packets will be meaningless to anyone who tries to read them. Remember, you can check to see if a website encrypts your data before transmission by looking to see if the URL uses HTTPS. It is also often possible, in your web browser settings, to set your web browser to only connect to websites that use encryption to transmit data.

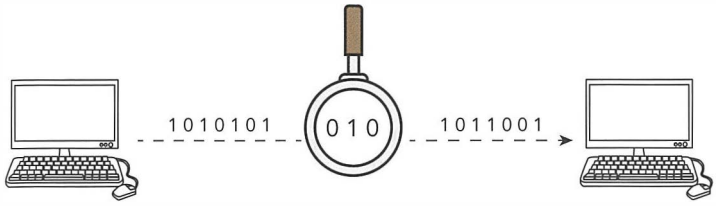


Figure 5.19: If your data is intercepted by someone, will it be encrypted? Check if the URL uses HTTPS

Have you ever gone into a café or a store and thought, 'Excellent, free Wi-Fi!' and you immediately connect to it? Perpetrators can also use this kind of incentive to get you to connect to the network through a system of theirs. Any data that you send using this connection can be intercepted by the perpetrator. It is advisable, if you see a free Wi-Fi connection, to ask the café or store if it is their free Wi-Fi before you connect to it.

In summary:

What happens?	Data packets are intercepted and examined through the use of software such as a packet sniffer.
What is the aim?	To steal your precious personal data for criminal activity such as identity theft or fraud.
What security solutions can be used?	Encryption such as the SSL protocol and visually checking the URL of a website to see if it uses the HTTPS protocol.

ACTIVITY 5.5

Use the internet to find out how a virtual private network (VPN) can be used to help prevent data interception.

Distributed denial of service (DDoS) attack

To carry out a **DDoS attack**, a perpetrator will first create a **botnet** (bot network, see Figure 5.20). This is done by sending **malware** to many computers to try and get you to

KEY WORDS

distributed denial of service (DDoS) attack: a type of cyber threat that targets a web server to cause it to crash and prevent access to the web pages that it stores.

botnet: a network of bots that are created to carry out a DDoS attack.

malware: a type of malicious software that is downloaded onto a user's computer or device.

accidentally download it. Once you have downloaded the malware, the computer can then be used as a **bot** by the perpetrator. Whilst the bot is not being used it is often called a **zombie**. Once the perpetrator wants to carry out the attack, they 'wake up' all the zombies to be used as bots. Each bot will begin sending multiple requests to access a web server. They are all set to send requests to the same web server all at the same time. This floods the web server with too many requests and it begins to struggle to deal with each request. The web server can only deal with a certain number of requests at a time and if it is sent too many it will begin to slow down and eventually crash. This will mean that users can no longer access the website. This causes a denial of service to any normal requests to access the web server. This type of attack is normally carried out on a web server, but it could be used to target other types of server and network hardware.

There are several reasons a perpetrator may carry out a DDoS attack. They may make a demand for money to the company to get them to pay to stop the attack from happening. They may want to punish the company for something that they perpetrator thinks they have done wrong, such as an unethical issue. This would be in an act of revenge or activism.

The main security solution that a company can use to stop a DDoS attack occurring on their web server is to use a **proxy server**. A proxy server can be used as a barrier to a web server by getting all the requests for the web server to pass through the proxy server first. The proxy server can examine each request and forward it on to the web server to allow access. If the proxy server begins to receive many requests, it can begin to allow them to pass at a slower rate to avoid the web server being flooded with all the requests at the same time. This may mean access to the website is slower, but still possible by users that want to legitimately access it. A proxy server can also use a feature called caching. This feature can help prevent a DDoS from being successful as once the proxy server has received a request the access the website from a certain IP address, it will stop any repeated requests for a short period of time from the same IP address. Therefore, if a bot is sending multiple requests, not all of them will be passed onto the web server as some will be stopped by the proxy server. This makes it more difficult for a perpetrator to flood the web server.

As a user, you can try and prevent your computer ever being used as a bot in a DDoS attack by regularly scanning your computer with **anti-malware** software. This is a type of software that scans all of the files in your computer to see if any of them are known to be malware. It does this by comparing each file against a known list of malware to see if any of the files on your computer match. If it finds a file that matches, it will put it in a special area called a quarantine area. It will then inform you that it has found what it thinks is malware and will ask you if you want to delete the file. By scanning your computer with anti-malware software on a regular basis, you can help make sure that your computer is free from any malware that might be used to turn it into a bot in a DDoS attack.

KEY WORDS

bot: the name given to a computer that has had malware downloaded onto it that will cause it to be used in a DDoS attack.

zombie: the name given to a bot that has malware downloaded onto it to allow it to be used in a DDoS attack, but it currently isn't being used in this way. It is effectively a dormant or sleeping bot.

proxy server: a method of cyber security that examines requests sent by devices to access the web pages stored on a web server.

anti-malware: a type of software that scans a computer or device with the purpose of finding and removing malware.

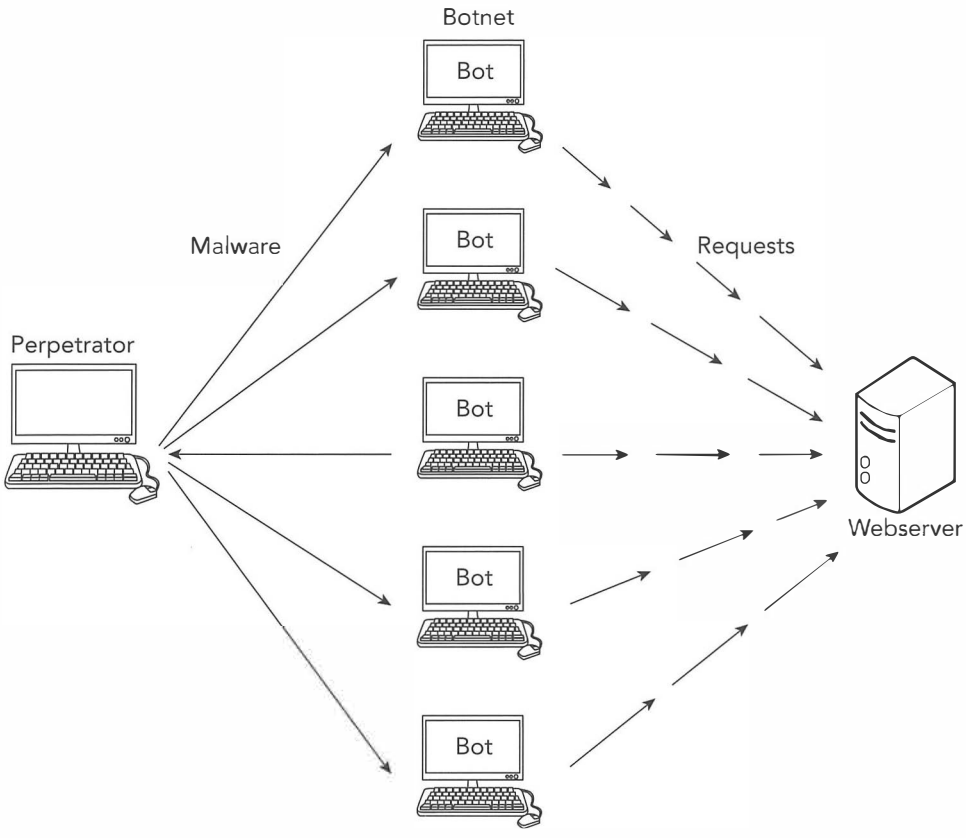


Figure 5.20: A botnet can cause a DDoS attack

In summary:

What happens?	Malware is sent to lots of computers to turn them into bots to create a botnet. The perpetrator then uses the botnet to send many requests to a web server, all at the same time.
What is the aim?	The aim is to cause the web server to crash. The perpetrator may be doing this to demand money for it to stop, or as an act of revenge or activism.
What security solutions can be used?	The company can use a proxy server to act as a barrier that filters requests to the web server. You can scan your computer with anti-malware on a regular basis to try and prevent it being used as a bot.

ACTIVITY 5.6

Use the internet to find out about a huge DDoS attack that was carried out on 21st October 2016, and which companies it affected.

Hacking

Hacking is the act of trying to gain unauthorised access to data. A perpetrator that does this is known as a **hacker**. They can do this by using a range of technical skills that allow them to break into a computer or a network. This can be done by going through methods such as a brute-force attack (this is sometimes known as cracking) or by exploiting vulnerabilities in a computer or a network, which are areas that the hacker can identify that lack the protection of security solutions.

Hackers will gain access to a system for several reasons. This could be to steal your data to try and steal your identity, as an act of revenge, to corrupt your data or to leak confidential data about a company or project, or an act of activism.

The main way that you can stop your computer being hacked is to have as many security solutions in place as possible. One of the main security solutions that can be used is a **firewall**. A firewall is a barrier that can be used between your data and the rest of a network. It can be software based or hardware based. Any data that is sent into and out of your computer is examined by the firewall. Rules can be set for this data by giving your firewall a set of criteria to examine the data against. The data that is sent through the firewall is compared to the criteria. The firewall can be set to accept the data and allow it to pass through the firewall, or reject the data and stop it from passing through the firewall, depending on whether it does or does not meet the criteria. This added security solution may help identify data that a hacker has sent that looks malicious and reject this data to stop the hacker getting into the computer or network. A firewall can also be used to close certain **ports** to a network. A hacker may look to find a vulnerability, such as an open port, to get into a network. A firewall can be set to close as many ports as it can that are not being used, to try and prevent a hacker exploiting an open port to gain access.

One kind of vulnerability that hackers may try to exploit is out of date software. Sometimes, a software company may find that they have an issue with their software that could allow a hacker to gain unauthorised access. If you have downloaded this software, you have also created this vulnerability on your computer. As software companies become aware of these issues, they develop their software and issue updates to protect against hackers. It is therefore vital that you update your software on a regular basis when software updates are released. This helps limit any vulnerabilities you have on your computer that a hacker can exploit. A way to make sure that your software is always up to date is to set your software to automatically update. This means that as soon as a software update is released, your computer will automatically download the update and you don't need to remember to do this.

You can also password protect all your data with strong or biometric passwords to help stop a hacker gaining unauthorised access to your data. And you can add two-step verification to all your accounts.



Figure 5.21: Good security will protect computers against hackers

In summary:

What happens?	A hacker gains unauthorised access to your data by exploiting a vulnerability in your computer or network.
What is the aim?	The aim is to steal, corrupt or leak your data for criminal activity.
What security solutions can be used?	A firewall, automatic software updates, strong or biometric passwords and two-step verification.

KEY WORDS

hacking: the act of gaining unauthorised access to data. This is normally done to steal or corrupt the data.

hacker: the name given to a perpetrator that carries out an act of hacking.

firewall: a cyber security method that is used to examine incoming and outgoing traffic from a computer or network.

port: an entry point into a computer or network.

Questions

9 What is a hacker?

10 How can you prevent a hacker gaining access to your data?

Malware

Malware is a term used to describe any malicious software that is designed to disrupt your computer or data. There are several types of malicious software that you need to know about: **virus**, **worm**, **trojan horse**, **spyware**, **adware** and **ransomware**. See Table 5.1.

Malware	Description
Virus	This is a computer program that is downloaded onto your hard drive. Once it has downloaded it replicates itself and corrupts your stored data or uses up all the available memory in your computer, causing it to slow down and crash.
Worm	This is a computer program that also replicates itself. It has a different aim to a virus though. It looks to find vulnerability holes in a network to use to replicate itself. In doing this it will clog up the bandwidth of a network and slow the network down.
Spyware	This is a computer program that is downloaded onto your hard drive and is designed to record your actions on your computer. A common form of spyware is a keylogger. This records all the key presses that a you carry out on your keyboard. This data is then sent to the perpetrator where it can be analysed to identify patterns in your data. Patterns in your data could reveal personal data such as your passwords. This can allow access to your online accounts for fraud and identity theft.
Trojan horse	This is a computer program that is used to disguise other malware. It is designed to look like a harmless piece of software, such as an application or game, but it contains other malware such as a virus or spyware. Once the trojan horse is downloaded onto your hard drive, it releases the other malware that it contains.
Adware	This is a computer program that is designed to automatically created pop up and banner adverts when you are online. These adverts are often unwanted and can be irritating and frustrating. The adware creators are given money by the companies that appear in the advertisements when you click on them.
Ransomware	This is a computer program that is downloaded onto your hard drive or other hardware. It is designed to encrypt your data and stop you from gaining access to it. The creators of the ransomware will demand a ransom (a set amount of money) for the data to be decrypted. They will often threaten to release and leak all your stored data if the ransom is not paid.

Table 5.1: Different types of malware

KEY WORDS

virus: a software based cyber threat that replicates itself with the aim of corrupting data or filling up the available memory space in a computer, causing it to crash.

worm: a software based cyber threat that replicates itself across a network using vulnerabilities that it finds, with the aim of clogging up the bandwidth.

trojan horse: a software based cyber threat that is used to disguise other malware to try and smuggle it into a computer or network.

spyware: a software based cyber threat that spies on a user's action whilst using a computer, such as logging their key presses.

adware: a software based cyber issue that automatically creates popup advertisements.

ransomware: a software based cyber threat that encrypts a user's data to stop them gaining access to it until a ransom (money) is paid.

There are several security solutions that you can use to protect against the threat of the different types of malware.

An **anti-virus** software can be used to scan your computer for viruses. This is a type of software that scans all of the files in your computer to see if any of them are known to be a virus. It does this by comparing each file against a known list of viruses to see if any of the files on your computer match. If it finds a file that matches, it will put it in a special area called a quarantine area. It will then inform you that it has found what it thinks is a virus and will ask you if you want to delete the file. By scanning your computer with anti-virus software on a regular basis, you can help make sure that your computer is free from any viruses. It is also important to update your anti-virus software. Perpetrators are constantly developing new viruses. The anti-virus software developers are constantly finding these and update their database used in the scan. You need to make sure that you are updating your anti-virus to make sure that the database is up to date and all the viruses are found.

KEY WORD

anti-virus: a type of software that scans a computer for viruses with the purpose of finding and removing them.



Figure 5.22: Anti-virus software should be kept up to date to find any and all viruses

Anti-malware software can be installed and used to scan your computer and network for other types of malware, such as worms, spyware, adware and ransomware. This operates in a similar way to the anti-virus software by scanning, quarantining and deleting any malware found. The anti-malware that looks specifically for spyware is called anti-spyware.

A firewall can be used to help prevent malware being downloaded onto your computer. The data coming into and out of your computer can be examined against the criteria set for the firewall. If the data looks malicious, the firewall will reject it and stop it from being downloaded onto your computer. It is useful that a firewall examines both the data incoming and outgoing data from your computer, especially in the prevention of spyware. As the firewall is examining the data leaving your computer as well, it may recognise a malicious program that is trying to leave your computer that contains all the keypresses that you have made on your keyboard. If the firewall rejects this data, it will not be able to leave your computer, preventing it being sent to the perpetrator to be analysed.

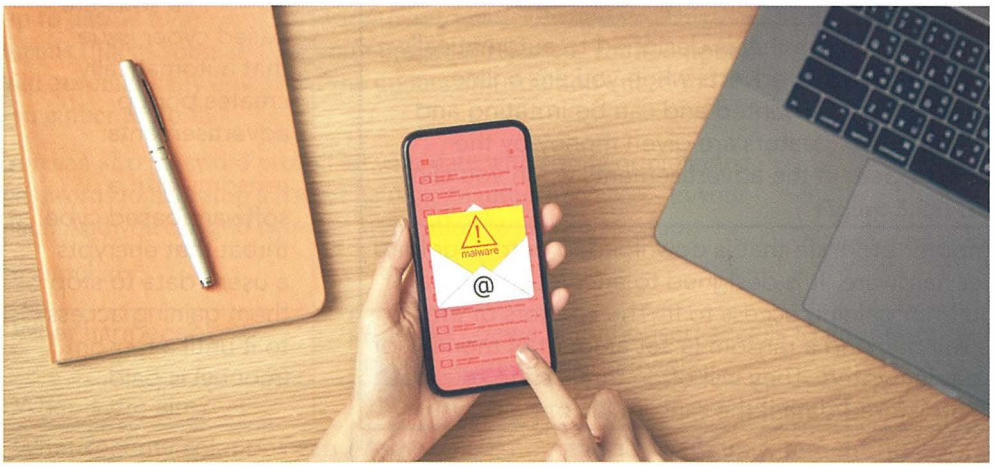


Figure 5.23: Malware can be sent to a computer via email

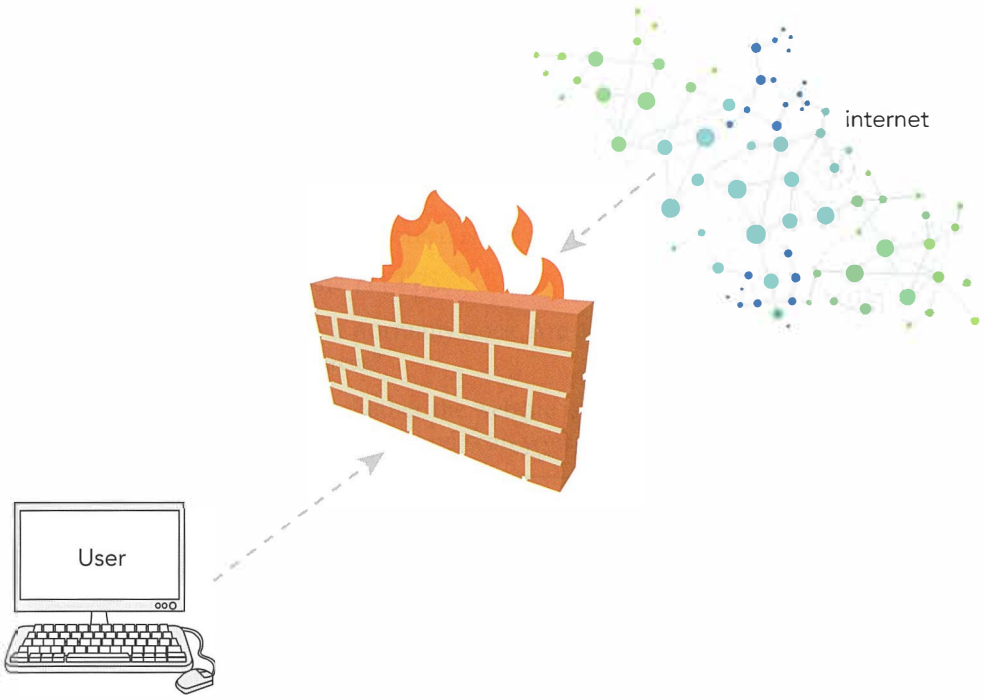


Figure 5.24: A firewall protects a user's computer from security threats on internet

You can make sure that you are very cautious when clicking any links and downloading any software online. Malware is often hidden in what looks like an innocent piece of software or link, for example, a game that looks enticing to play. There could be malware hidden inside a trojan horse in the game that may not be detected by a firewall. If you download the game, it will download the trojan horse, allowing other malware to be released onto your computer. You should only ever download software from companies and sources that you know can be trusted, no matter how fun or enticing it may look.

You can also store a **backup** of your data that is not attached to your computer or network on a constant basis. This means that if your data is damaged by malware, or encrypted using ransomware, you still have access to a copy of your data. This can prevent you needing to pay a ransom to gain access to your data again, but it will not stop the perpetrator from leaking your data. Therefore, it is advised not to keep any personal data on your computer that could be accessed and leaked by a perpetrator of ransomware. If you do keep personal data stored on your computer, you should encrypt the data yourself, so if the perpetrator did gain access to the data, it would be meaningless to them and they wouldn't be able to leak it.

To avoid the risk of personal data, such as passwords, being sent to a perpetrator, you could use an onscreen virtual keyboard rather than a physical one to enter passwords. This means that you don't need to make any key presses, so these are not recorded for your password, so any data that was sent to a perpetrator would not contain your passwords. Some companies also add drop down boxes for you to select characters from to input your password. They may also ask you for random characters from your password, and not your whole password. This is to prevent you needing to enter your full password and to stop your full password being sent to the perpetrator.

KEY WORD

data backup: a copy of data that is stored separate from the computer.

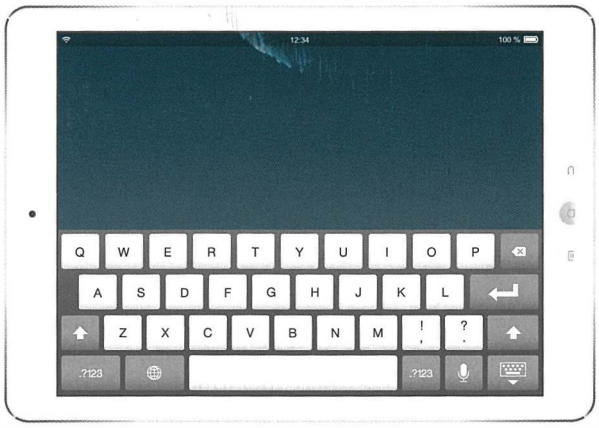


Figure 5.25: An onscreen virtual keyboard

In summary:

What happens?	Malicious software is downloaded onto your computer or network.
What is the aim?	Malicious software is used to corrupt your data, gain access to your data or damage your hardware.
What security solutions can be used?	Anti-malware software, including anti-virus and anti-spyware, a firewall, exercising caution when downloading software and creating a backup of data.

Pharming

Pharming is used by perpetrators to try and get your personal data, such as your username and password for your online accounts. They can then log into the account, pretending to be you, and use your account to buy products online with your money. It can also be used to get more personal details from you, such as your address, that can be used by the perpetrator for identity theft and fraud.

The perpetrator will try and entice you into clicking a link or icon to start a download. This could be done in various ways including sending an email, creating a pop up advert or just having a link available on a web page. When you click the link, a download will begin and as part of that download, malicious software will be downloaded onto your **hard drive**. This malicious software is designed to redirect certain website requests you have to go to a fake website instead (see Figure 5.26). The perpetrator will set up a fake website to look very similar to a genuine website, such as PayPal or your social media page. When you next type in the URL to request the web page, the malicious code that has been downloaded on your hard drive will redirect the request to the fake website instead. The perpetrator will hope that you think you have opened the genuine website and when you type in your login details, these will be sent to the perpetrator instead.

If you notice that the website is fake after typing in your login detail, it is very important that you change your password for that website immediately. This can help stop the perpetrator being able to log into your account.

As pharming uses malicious code that is installed on your hard drive, you can use anti-malware software to remove it. The anti-malware software will scan the files stored on

KEY WORDS

pharming: a type of cyber threat that involves downloading malicious software onto a user's hard drive, that redirects a request aimed at a genuine website to a fake website instead. The fake website will encourage the user to input their personal data, allowing it to be stolen.

hard drive: a storage device that stores data permanently, until it is deleted by the user.

your computer and will help to find the file that is redirecting you to the fake website. The anti-malware software will quarantine this file and let you delete it. When you type in the URL now the redirection should stop and you should be taken to the genuine website.

One thing that you can do to help prevent pharming is to only download software or click links that you know are from a trusted source. You should also thoroughly check any website before you enter your personal details. You can look to see:

- If the URL at the top of the screen is the correct one.
- If the site is secured using HTTPS.
- If the website looks like it normally does. Are all the usual colours, images and text there in the same places?

If you are in any doubt at all whether the website is genuine, close the website without entering any of your details and scan your computer immediately with anti-malware software.

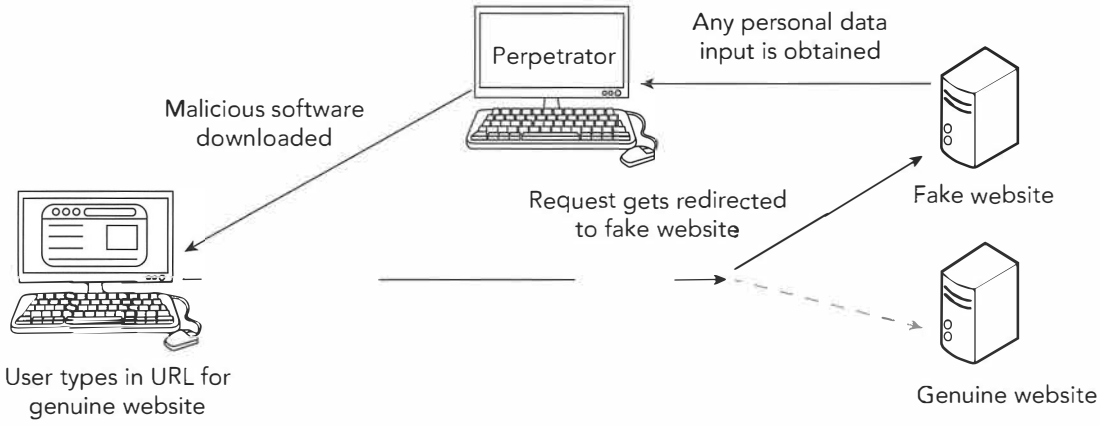


Figure 5.26: Pharming is a cyber threat where fake websites are used to steal data

In summary:

What happens?	Malicious software is downloaded onto your hard drive that will redirect you to a fake web page when you input the genuine URL.
What is the aim?	To get your personal data to commit criminal activity such as stealing your money, identity theft or fraud.
What security solutions can be used?	Anti-malware software, visually checking the web page and only downloading data from trusted sources.

Question

11 What can you do to prevent a perpetrator getting your personal data through pharming?

Phishing

Phishing is also used by perpetrators to try and get your personal data, such as your username and password for your online accounts. They can then log into the account,

KEY WORD

phishing: a type of cyber threat that involves sending a user a fake email that is designed to look genuine. It will encourage the user to provide their personal data either by clicking a link to a fake website, or by responding to the email.

pretending to be you and use your account to buy products online with your money. It can also be used to get more personal details from you, such as your address, that can be used by the perpetrator for identity theft and fraud.

The perpetrator will again try and entice you into clicking a link. In phishing, this is done by sending you an email. The perpetrator will create an email that looks like it is from a genuine company, such as your bank. The email will try and get you to click a link that will take you to a website. The email will try and get you to click the link by telling you things like:

- It looks like there has been unusual activity on your account, please click the link and log in to check.
- You have won a prize, please click the link and enter your details to claim the prize.
- Confirmation is needed that you have made this purchase, please click the link and log into your account to confirm whether or not you have made this purchase.
- You are being paid a refund on your taxes, please click the link and enter your credit card details so that you can claim the refund.

The link will take you to website that will not be the genuine website, it will be a fake website that has been made to look like the genuine one. If you then input your personal details these will be sent to the perpetrator and they will be able to access your account or try and build a profile to steal your identity.

You can help prevent yourself getting fooled by any phishing scams by checking:

- The spelling and tone of the text in the email. Is everything spelt correctly? Is the email written in a professional tone? Most genuine companies will spend a lot of time making sure their emails are professionally written.
- Whether the email has been personalised or if it is addressed in a generic way, e.g. Dear customer. Many phishing emails are sent out in bulk and this often means they are not individually addressed.
- The URL that is attached to the link. If you hover your mouse pointer over the link, you should be able to see the URL that the link is connected to. Check if this is the correct URL for this company.
- If the URL at the top of the screen is the correct one.
- If the site is secured using HTTPS.
- If the website looks like it normally does. Are all the usual colours, images and text there in the same places?

Most companies will not ask you to provide your personal details over the internet in this way. If you are in any doubt at all whether the website is genuine, close the website without entering any of your details. If you have entered any of your details and realise after that you think it is a phishing scam, make sure that you change the password to your account and any other accounts that use that password immediately.



Figure 5.27: In a phishing scam perpetrators will try to get your password

Question

12 What are three things that you can look for to identify if an email is a phishing scam?

Social engineering

Social engineering relies on manipulating and deceiving people into providing confidential or personal data that can be used in criminal activity such as identity theft and fraud, and hacking into computers systems or networks. Social engineering can take many different forms. Phishing is also one form of social engineering, in that it manipulates you into providing your personal data.

KEY WORD

social engineering: a cyber threat that involves manipulating or deceiving people into providing confidential or personal data.

Have you ever seen one of those quizzes on social media that ask you to provide lots of fun details about you, like your favourite colour, the name of your pet and your favourite food? These quizzes are often created by perpetrators trying to find out key information about you. Many people base their passwords around this type of information, so a perpetrator can gather lot of information like this about you every time you fill in one of these quizzes on your social media. They can then use this data along with a brute-force password program to try and get into your online accounts. Advice for the future, do not fill out these quizzes! This is another form of social engineering, it manipulates and deceives you into thinking you are filling out a fun quiz to learn more about your friends, but you are actually providing key information that a perpetrator can use against you. One way that you can limit a perpetrator's access to your data on social media is to have your privacy settings set to the highest level of privacy. This will normally mean that only people that you are connected to will be allowed to see your data.

Social engineering can be used to target companies for their data too. A perpetrator could make contact with the company and try and manipulate and deceive employees into providing key information that can be used to hack the company systems. Here are two examples of how this could be done:

Example 1

Large companies often have quite a large IT department that are often not too well known by all the employees. It is a common occurrence that an employee may have an IT issue with their computer in a morning. They will normally report this issue to the IT department in their company and will be told someone will contact them from the department, shortly. Therefore, a perpetrator can try and use this situation. They get hold of a list of all the telephone numbers of the employees in the company and begin to call them. They will say that they are from the IT department in the company and say that they are aware of the IT issue that they have reported. Most employees will probably just say they haven't reported an issue, but if they happen to call an employee that has, that employee will think it is the IT department calling to sort their problem. The perpetrator will then say that they just need the employees login details to access their system to see what is wrong with it. The employee may be tricked into providing their login details and

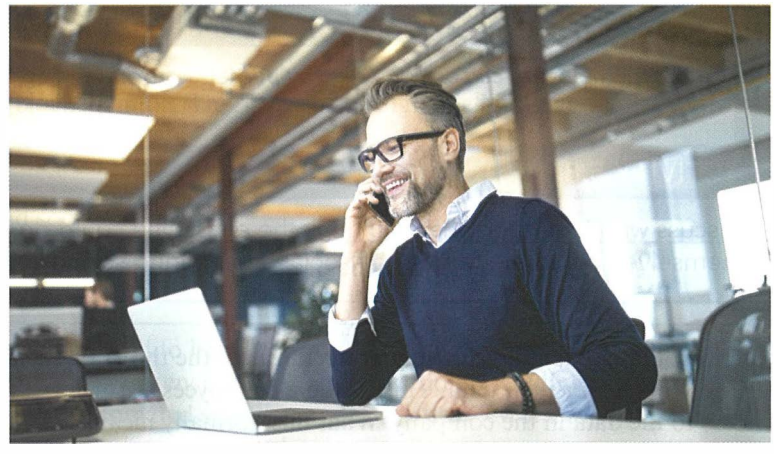


Figure 5.28: Employees of large companies can be contacted by perpetrators claiming to be from the IT department

now the perpetrator can get into the computer systems of the business and steal any data that they think could be useful to them, or look to introduce malware.

ACTIVITY 5.7

Discuss with a partner what you think the employee could do to stop themselves being the target of a social engineering scam like this one.

Example 2

Employees often go for dinner together or out for a relaxing coffee after work on a Friday. A perpetrator finds out where employees do this for the company that they want to target. The perpetrator places themselves close to where the employees will normally sit. They will start a conversation with the employees and become friendly with them. They will start to ask lots of questions about the company an employee works for. The employee may mistake this as the person taking an interest in their life. Over a few what seem like accidental meetings with the employee the perpetrator has asked lots of questions that has allowed them to build up a profile about the company, such as information about where they store their data and what kind of security systems they have. The perpetrator can use this information to hack into the company systems and network.



Figure 5.29: Employees can be contacted by perpetrators pretending to be colleagues

ACTIVITY 5.8

Discuss with a partner what you think the employee could do to stop themselves being the target of a social engineering scam like this one.

One way that companies can help protect against the threat of social engineering is to use **access level** for their data. This is where employees are given different levels of access to the data in the company. What this normally means is that an employee will only have access to the data that they directly use on a daily basis. It also often means that the only people that have access to more confidential data are the more senior employees in the company. Levels of access are linked to an employee username.

KEY WORD

access level: the amount of direct access a user is given to data and resources. This could be set to be only certain sections of a whole collection of data and resources.

REFLECTION

How did you come to a decision about what the employees could have done in Activities 5.7 and 5.8?

The **username** will have data connected to it that will show the computer system which data the employee is allowed access to. This means that if an employee is targeted by a social engineering scam and fooled by it, the data that the perpetrator can gain access to may be limited. A company can also train employees to recognise social engineering to make them more aware of when it might be happening to them.

In summary:

What happens?	A perpetrator will try and manipulate and deceive you into providing them with personal or confidential data.
What is the aim?	To get data to commit criminal activity, such as stealing your money, hacking into a computer network and identity theft or fraud.
What security solutions can be used?	Access level and awareness of how social engineering is conducted.

KEY WORD

username: a text-based identification that is given to a user that identifies the level of access that they have to stored data and resources.

SUMMARY

The internet is the infrastructure that is used to access the world wide web.
The world wide web is the collection of all the web pages that are available.
A URL is a text based address for a web page that contains the protocol, the domain name and the web page name or file name.
The HTTP protocol is used to transmit data between a computer and a web server. This data is not encrypted when sent using this protocol.
The HTTPS protocol is the encrypted version of the HTTP protocol. It uses the SSL protocol to encrypt the data.
The main purpose of a web browser is to render HTML to allow a user to view web pages. It also has other functions such as recording a user's history and storing bookmarks, favourites and cookies.
Cookies are used to save your personal data and track your online preferences. They can be session or persistent based.
Web pages are located, retrieved and displayed using a web browser, a DNS and a web server.
A digital currency is a currency that only exists electronically. An example can be cryptocurrency and this kind of digital currency is tracked using blockchaining. This is a type of digital ledger that records all payments made using the currency.
There are several cyber security threats that exist such as brute-force attacks, data interception, DDoS, hacking, malware, pharming, phishing and social engineering.
The aim of most of the cyber security threats is to steal your personal data to commit identity theft and fraud, or as an act of revenge or activism.
There are a range of cyber security solutions that can be used to help keep your data safe from security threats such as anti-malware software, firewalls, proxy servers, encryption, authentication (including biometric passwords and two-step verification), privacy settings, automated software updates, access levels and visual checks such as checking the URL that is connected to a link or download.

EXAM-STYLE QUESTIONS

1 **State** the name given to the infrastructure that allows a user to access the world wide web. [1]

2 A URL is the name given to the text based address for a web page. The web page name is one part that is used to create the URL.

a **Identify** the name for *one* other part of a URL. [1]

b State the name of the hardware that finds the corresponding IP address for a URL. [1]

[Total: 2]

3 The given paragraph describes how a web page is located and retrieved to be displayed on a user's device. Complete the missing words in the paragraph. [5]

The user types the URL into the address bar in the ..... This is sent to the ..... where the corresponding IP address is found. The IP address is sent to the web browser. The web browser uses the IP address to send a request to the ..... where the web pages are stored using the ..... protocol. The web pages are sent to the web browser. The web browser renders the ..... to display the web page.

4 One function of a web browser is to store cookies.

a Identify *two* other functions of a web browser. [2]

b State the name of the type of cookie that is stored in a computer until it is deleted by the user or by the web browser when it expires. [1]

[Total: 3]

5 **Give** *one* benefit of using blockchain to track a digital currency. [1]

6 A website company is worried about a distributed denial of service (DDoS) attack being carried out on their web server.

a State the name of a security solution that can be used to help prevent a DDoS attack taking place. [1]

b In a DDoS attack, a perpetrator creates a botnet.

Describe how the botnet is created by the perpetrator. [3]

[Total: 4]

7 Draw a diagram to show how data interception is carried out. [3]

COMMAND WORDS

state: express in clear terms.

identify: name / select / recognise

COMMAND WORDS

give: produce an answer from a given source or recall / memory.

describe: state the points of a topic / give characteristics and main features.

CONTINUED

8 Draw a line to connect the security solution to the correct description. **[3]**

Security solution	Description
Firewall	This is a protocol that encrypts data that is sent across the internet.
Anti-malware	This is software that can be used to scan a computer for malicious files.
Access levels	This can be hardware or software based and it examines traffic incoming and outgoing from a computer system or network.
SSL	This is connected to a user's username and provides information about what data the user is allowed to access on a system.

9 A student finds that a perpetrator has managed to hack into their social media account.

Explain *two* ways the perpetrator could have managed to do this. **[4]**

COMMAND WORD

explain: set out purposes or reasons / make the relationships between things evident / provide why and/or how and support with relevant evidence.

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics.
This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising.
You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
explain the difference between the internet and the world wide web.	5.1			
explain what is meant by a URL and what the different parts of it are.	5.2			
describe how data is sent using both the HTTP and HTTPS protocols.	5.2			
describe how web pages are located, retrieved and displayed using a web browser, DNS and a web server.	5.2			
explain the main purpose of a web browser and I can identify other functions that they often have.	5.2			
explain how cookies are used and what the difference is between a session cookie and a persistent cookie.	5.3			

CONTINUED

I can...	See topic	Needs more work	Getting there	Confident to move on
explain what is meant by a digital currency.	5.4			
explain how blockchaining is used to track certain digital currencies.	5.4			
describe how a brute-force attack is carried out, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			
describe how data can be intercepted, what the aim of it is and how to help prevent it happening.	5.5			
describe how a DDoS is carried out, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			
describe what is meant by hacking, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			
describe what is meant by a virus, what the aim of it is and what security solutions can be used to help prevent it being downloaded.	5.5			
describe what is meant by a worm, what the aim of it is and what security solutions can be used to help prevent it being downloaded.	5.5			
describe what is meant by a trojan horse, what the aim of it is and what security solutions can be used to help prevent it being downloaded.	5.5			
describe what is meant by spyware, what the aim of it is and what security solutions can be used to help prevent it being downloaded.	5.5			
describe what is meant by adware, what the aim of it is and what security solutions can be used to help prevent it being effective.	5.5			
describe what is meant by pharming, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			
describe what is meant by phishing, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			
describe what is meant by social engineering, what the aim of it is and what security solutions can be used to help prevent it happening.	5.5			